Why Enterprises Pay the Frontier Premium REV 2.2

October 2026 update · five mechanisms behind the closed-vs-open procurement decision · modelled economics

∑ **Token price is the wrong denominator.** Cost per unit of output is **(1 + f) / m** — model spend is *additive*, throughput is a *divisor*. At a revised **2.4–5.8% of loaded labour**, the frontier needs only a **4.0% throughput edge** over open weights to break even — a bar that p^N compounding clears at any horizon N ≥ 2. The genuine overpayment isn't choosing proprietary; it's **routing everything to a flagship max-effort tier**.

STRUCTURED ABSTRACT

QUESTION	Why do large enterprises pay a substantial premium for frontier proprietary models when open weights reach roughly 79% of composite benchmark capability (87% in June 2026) at a fraction of the token price?
FINDING	Token price is the wrong denominator. Cost per unit of output is (1 + f) / m: model spend enters additively while throughput divides. At realistic spend the frontier needs only a ~4% throughput edge to break even. A two-point per-step reliability edge (97%→99%) clears that bar at any horizon N ≥ 2, and by a widening margin as N grows.
METHOD	Derived arithmetic (p^N compounding; the unit-cost identity and its isoclines) combined with parameterised models for self-hosting TCO and threshold-gated value. Benchmark figures are third-party reported; per-step reliability values are illustrative parameters, not measurements (Note 2).
SCOPE	Applies to knowledge-work and agentic software deployment at enterprise scale. Does not address consumer products, on-device inference, or research settings.

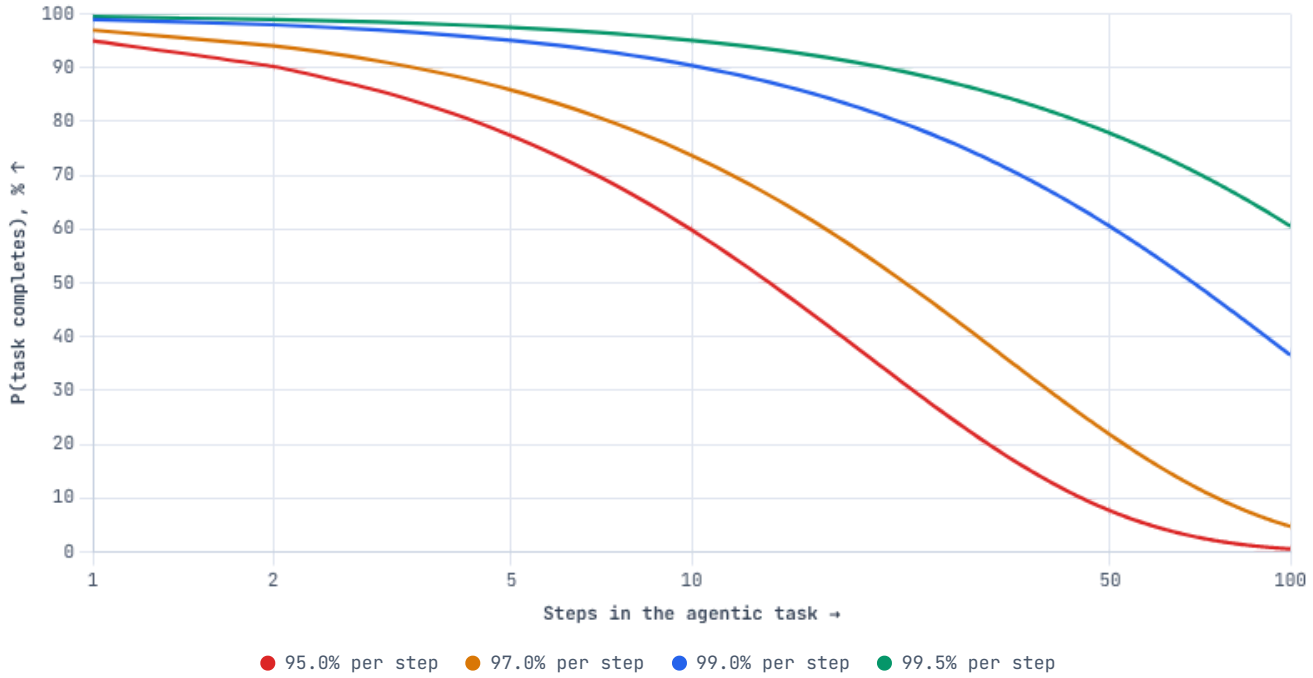
1 · Agentic error compounding

Task completion probability vs horizon · P(success) ≈ p^N

The single largest driver, and the one invisible on single-turn benchmarks. Success on an N-step agentic task approximates **p^N**. A 2-point per-step reliability edge is nearly invisible in a benchmark table and **decisive** by step 50.

Task completion probability vs agentic horizon

Y = P(all steps succeed), % · X = steps in the task (log scale)



PER-STEP RELIABILITY	10 STEPS	25 STEPS	50 STEPS	100 STEPS
95.0%	59.9%	27.7%	7.7%	0.6%
97.0%	73.7%	46.7%	21.8%	4.8%
99.0%	90.4%	77.8%	60.5%	36.6%
99.5%	95.1%	88.2%	77.8%	60.6%

WHY THIS DOMINATES

Moving per-step reliability from **97% to 99%** — a two-point delta a composite index would round away — is a **2.8× swing in completion at N=50** and **7.7× at N=100**. That is the boundary between "the agent ships the PR" and "an engineer babysits and re-drives." Read against driver 2: the completion-ratio gain from that edge is **2.1% at N=1, 4.2% at N=2, 22.6% at N=10** — it clears the 4.0% break-even bar from N=2 onward, by a widening margin. The bridge from completion probability to m is **accepted output** (limit 4 in driver 2): a task that fails to complete ships no feature and adds nothing to m.

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2 · The labour denominator

Unit cost = (1 + f) / m · f = spend as a fraction of loaded labour, m = throughput multiplier

REVISED REFRAMED

Cost per unit of output is **`(1 + f) / m`**, where f is model spend as a fraction of loaded labour and m is the throughput multiplier. **f is additive; m is a divisor.** A small numerator penalty buys a large denominator gain, which is why the frontier wins while spending several times the tokens.

Reading the formula

1

One seat, one year. Start with what a person costs you.

$250,000 loaded salary

+ f

Add what their AI costs, expressed as a fraction of that salary. f = 0.05 means you are now paying 1.05 salaries for that seat.

f = model spend ÷ salary

÷ m

Divide by how much more work the seat now produces. m = 1.45 means 1.45× the output.

m = throughput multiplier

=

What one unit of work now costs you, relative to before. Below 1.00 means cheaper.

(1 + f) / m

Worked example · one engineer, one year

Illustrative units of completed work — not a measurement

LINE	BASELINE	OPEN WEIGHTS	FRONTIER
Loaded salary	$250,000	$250,000	$250,000
Model spend	$0	$2,500/yr	$12,500/yr
as fraction of salary (f)	0%	1%	5%
Total cost of the seat	$250,000	$252,500	$262,500
Features shipped in the year	40	48	58
throughput multiplier (m)	1.00×	1.20×	1.45×
Cost per feature	**$6,250**	**$5,260**	**$4,526**
Unit cost index · (1 + f) / m	**1.000**	**0.842**	**0.724**

WHAT THE EXAMPLE SHOWS

The frontier seat costs 5% more and produces 45% more, so each feature lands at **72.4% of baseline cost** — and **14.0% below the open-weights seat**. Break-even against open weights needed only m ≥ 1.25 (about 50 features). It delivered 58. Feature counts are illustrative units of completed work, not a measurement.

Why the asymmetry is the whole argument

Elasticity of unit cost with respect to each input

INPUT	ELASTICITY	AT F = 5%	READING
Model spend · f	f / (1 + f)	**+0.048**	Nearly inelastic. Doubling the AI bill raises unit cost ~4.8%.
Throughput · m	-1 (exactly, always)	**-1.000**	Unit-elastic. A 10% throughput gain cuts unit cost 9.1%.

LEVERAGE

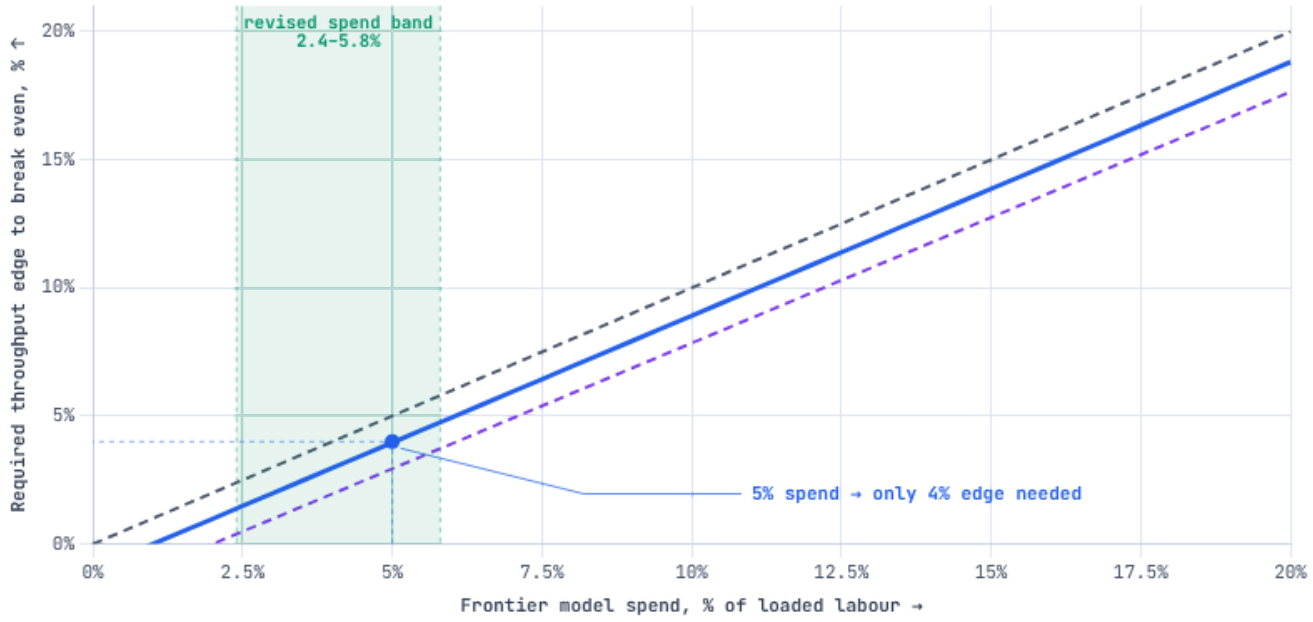
Percent change in unit cost per percent change in the input, evaluated at f = 5%. Throughput carries roughly **21× the leverage of spend**. The elasticity in f is self-limiting: spend matters more the more you spend, but only linearly and from a very low base.

Scenario comparison

SCENARIO	F · SPEND	M · THROUGHPUT	UNIT COST	VS BASELINE
Baseline · no tooling	0%	1.00×	**1.000**	—
Open weights	1%	1.20×	**0.842**	-15.8%
Frontier	5%	1.45×	**0.724**	-27.6%
Frontier · heavy agentic	10%	1.55×	**0.710**	-29.0%

Throughput edge the frontier must deliver to break even

Y = required advantage over open weights, % · X = frontier model spend as % of loaded labour · each curve is a different open-weights spend baseline · shaded band = revised realistic spend



BREAK-EVEN AT 5% SPEND
+4% throughput edge

BREAK-EVEN AT 10% SPEND
+8.9% throughput edge

BREAK-EVEN AT 15% SPEND
+13.9% throughput edge

FRONTIER UNIT COST
0.724 vs 0.842 open

BREAK-EVEN IN CLOSED FORM

Frontier beats open weights when m_F / m_O ≥ (1 + f_F) / (1 + f_O). Rearranged, the required throughput edge is **E = (f_F - f_O) / (1 + f_O)**. Two consequences: **E does not depend on m at all**, so the multipliers used throughout this document set the size of the win but never whether there is one; and for small f_O, **E ≈ Δf** — the extra spend expressed as a percentage of salary. In plain terms: if the frontier model costs four more points of a salary, it has to make that person 4% more productive.

What the identity cannot see

1 **Output must be homogeneous**

Q is “units of completed work.” Where the frontier model does things the alternative cannot do at all, m is undefined — that is a capability discontinuity, and driver 4 governs, not this identity.

2 **m is a workload-weighted average**

Real m is high on refactoring and boilerplate, low on novel design, and plausibly below 1 where review overhead exceeds generation savings. A composite m hides that spread — the same averaging trap criticised in driver 4.

3 **Fixed costs are omitted**

Integration, eval harnesses, training, and change management are absent. The fuller form is (1 + f + k) / m, where k is amortised adoption cost as a fraction of salary. In year one k can exceed f outright, which is why pilots underperform the model.

4 **Quality is invisible**

Two outputs count as one unit each regardless of defect rate. Strictly, m should be accepted output, not generated output. Driver 1 exists because this identity cannot see reliability.

5 **Loaded labour is assumed invariant**

If AI-augmented engineers command a wage premium, L rises and f falls mechanically — flattering the ratio for the wrong reason.

REVISED SPEND BAND — THE EARLIER 1–2% WAS ANCHORED TOO LOW

A saturated agentic engineer running parallel sessions lands nearer **$500–$1,200/seat/month**, or **2.4–5.8% of $250k loaded labour**, with a fat right tail *(MED — inferred from agentic usage patterns, not a surveyed figure)*. Raising f does not weaken the case: at 5% frontier spend against 1% open weights, break-even is a **4.0% throughput edge**. Even at 15% spend it is only 13.9%. The token-price argument was never load-bearing.

JEVONS DRIFT — AND WHERE THE FRAME BREAKS

As inference gets cheaper per unit of work, consumption rises faster than unit cost falls, so **total spend goes up** — Gartner predicted in June 2026 that AI coding costs will exceed the average developer salary by 2028. The 2.4–5.8% figure will drift upward, and that drift is a health signal rather than a cost problem. Past some point "tooling as a percentage of labour" stops being the right accounting frame entirely: compute becomes a **factor of production alongside labour** rather than an overhead line charged against it. That is when the procurement conversation changes shape.

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3 · Substitution modes

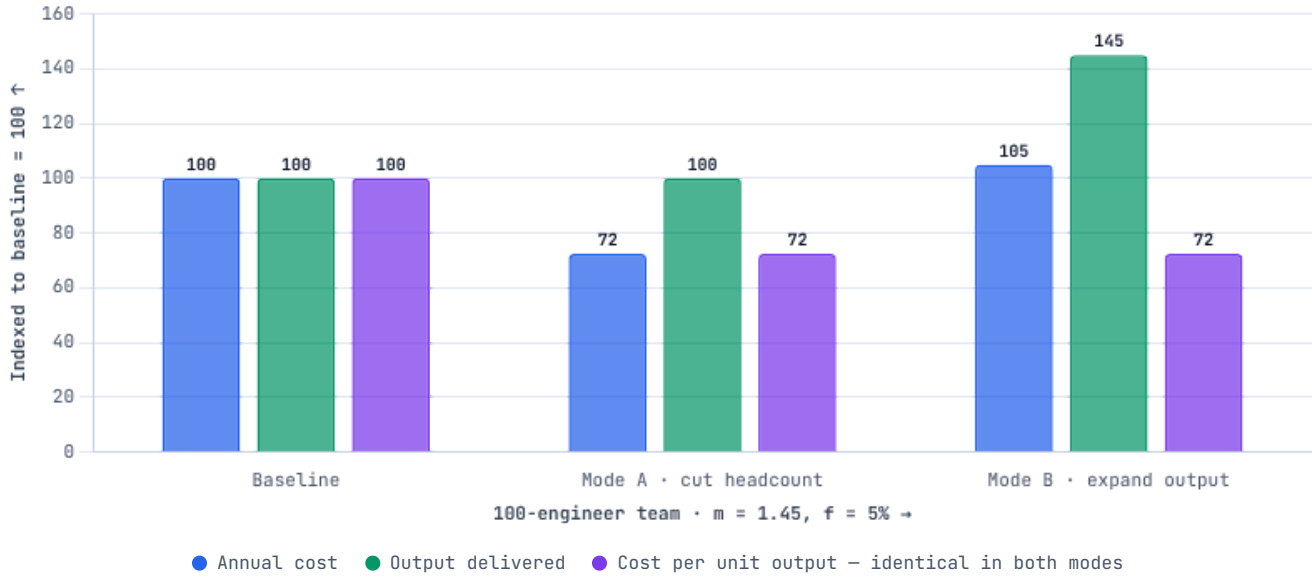
Two ways to bank the same multiplier · identical on unit cost, different on total value

NEW IN THIS EDITION

A throughput multiplier can be banked two ways: hold output and cut headcount, or hold headcount and expand output. They are **identical on unit cost and completely different on total value**. Illustration below: a 100-engineer team at m = 1.45, f = 5%.

Two ways to bank the same multiplier

Indexed to baseline = 100 · cost, output, and cost per unit of output



MODE A

Reduce headcount

Value is **capped at the labour line** — you recover at most the salaries eliminated, and the gain terminates. Note that **f itself does not move**: N/m engineers at the same per-seat spend scale numerator and denominator together. What shrinks is the *absolute* AI budget, which can make a high-leverage line item look small and invite under-investment — a perception risk, not arithmetic.

Forced when: **demand is the binding constraint**

MODE B

Hold headcount, expand output

Value is bounded by **opportunity set, not by cost**. Unbounded upside where a project backlog exists — additional revenue-generating work rather than a one-time cost recovery.

Available when: **engineering capacity is the constraint**

THE CHOICE IS A DIAGNOSTIC, NOT A PREFERENCE

Both modes land at **72.4 on cost per unit of output** — the arithmetic cannot distinguish them. What separates them is which constraint actually binds. Most software organisations claim engineering capacity is their limit, often accurately; firms that reflexively take Mode A are usually revealing that their real constraint was **never headcount**. That is a strategy disclosure, not a procurement decision.

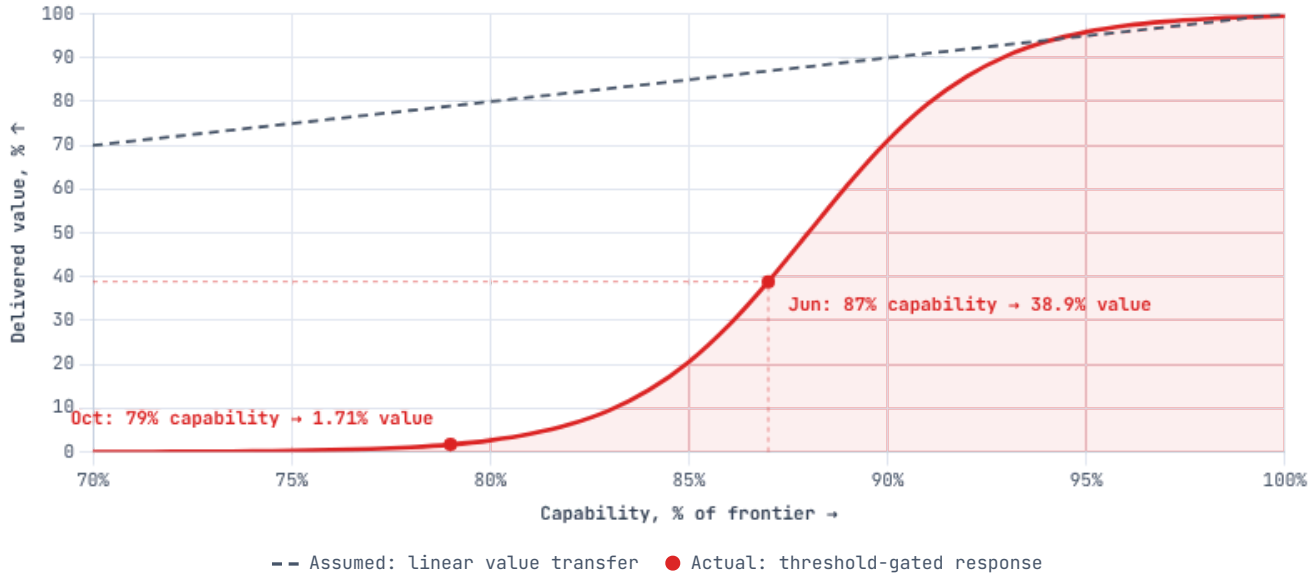
4 · Threshold-gated value

Delivered value vs capability · the composite-average trap, priced

"Open weights reach ~79% of proprietary" (87% in June) implies linear value transfer. On threshold-gated work — the task either clears the bar or it doesn't — **delivered value is sigmoid**, and the index points between open weights and the frontier straddle the knee.

Delivered value vs model capability

Illustrative: assumed linear transfer vs threshold-gated response · knee at ~88 · marks at the June (87%) and October (79%) gaps



THE COMPOSITE-AVERAGE TRAP, PRICED

On this curve June's **87% of frontier capability delivered ~38.9% of the value**, not 87%. October's wider gap, **79%, sits below the knee entirely at ~1.7%**: the exact figure is an artefact of the chosen curve, but the direction is not — a wider gap on gated work forfeits value far faster than the index suggests. The benchmark isn't wrong — a mean over ten evals is simply the wrong functional form for work gated on a pass threshold. The easy 80% of tasks were automatable two model generations ago; the residue is exactly what sits on the knee. *Illustrative model — LOW confidence on curvature, HIGH on direction.*

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5 · Self-hosting total cost of ownership

Effective cost per 1M output tokens vs fleet utilisation · 4× 8×H200 + 3 ops FTE

Open weights are free; **serving them is not**. Fixed cost is amortised over tokens actually produced, so effective unit cost is governed by utilisation — and enterprise diurnal load runs **5–10× peak-to-trough**.

Effective cost per 1M output tokens vs fleet utilisation

Modelled: 4× 8×H200 nodes + 3 ops FTE · shaded band = realistic enterprise utilisation



BREAK-EVEN VS OPEN-WEIGHTS API

none API cheaper at any load

BREAK-EVEN VS MID-TIER PROPRIETARY

34.3% utilisation

BREAK-EVEN VS FRONTIER FLAGSHIP

6.9% utilisation

FIXED COST MODELLED

$2.16M /yr · infra + ops

MODEL ASSUMPTIONS — LOAD-BEARING, TAGGED

Infra **$1.26M/yr** (4 nodes × $36/hr, 8×H200 rented at $4.50/GPU-hr, the low end of October 2026 on-demand pricing — *MED*) plus ops **$900k/yr** (3 FTE × $300k loaded — *MED*) = **$2.16M/yr fixed**. Peak capacity 5,000 output tok/s per node for a ~100B-class MoE under heavy batching — *LOW confidence; this is the swing variable*. Halve throughput and every crossover doubles. Reference prices are October 2026 list output prices. The qualitative result: **self-hosting beats a flagship price almost immediately, a mid-tier price only at the top of the realistic band, and a cheap open-weights API not at all** — open-weights API prices have fallen below what a private fleet can match, so the case for self-hosting is now privacy, residency, or control, not unit cost.

Net read

Eight drivers, with basis and confidence

Eight drivers, ordered by how much of the premium they explain, with basis and confidence on each.

DRIVER	MECHANISM	BASIS	CONF
Agentic error compounding	97%→99% per-step = 2.8× completion at N=50	Derived (p^N)	HIGH
Labour denominator	f is additive, m is a divisor — break-even edge only ~4%	Derived	HIGH
Substitution symmetry	Cut headcount and expand output share identical unit cost	Derived	HIGH
Threshold-gated value	79% composite capability (Oct 2026) → ~2% delivered value; 87% (Jun) → ~39%	Modelled	MED
Self-host TCO	At Oct 2026 prices, never undercuts a cheap open-weights API; beats mid-tier only at ≈34% utilisation	Modelled	LOW
Indemnity & compliance	IP indemnity, SOC 2 / HIPAA / FedRAMP, residency	Contractual	HIGH
Provenance & sanctions	Training-data lineage, backdoor risk, Entity List exposure	Qualitative	MED
Harness, not checkpoint	Caching, batch, tool-call reliability, agentic scaffolds	Qualitative	HIGH

LANE 1 · OPEN WEIGHTS

Bounded, high-volume, private

Classification, extraction, routing, embedding. Error-tolerant, short-horizon, and cheap enough per token that capability is not the constraint. **The open-weights case holds cleanly here** — usually through a hosted API, with self-hosting reserved for privacy and residency.

Also: **narrow fine-tunes** that beat general frontier on one task

LANE 2 · MID-TIER PROPRIETARY

General reasoning, good-enough

Where the **dominated-flagship trap** lives. A cost-tuned proprietary tier now scores above every open-weights model at a comparable cost per task (Oct 2026: 48 at $0.21 against the best open model's 46 at $0.13) — and at a fraction of a flagship's.

Trap: **max-effort by default** on work a mid tier clears

LANE 3 · FRONTIER

Long-horizon, calibration-critical

Agentic chains, must-not-hallucinate, threshold-gated. p^N and the sigmoid knee are both load-bearing. **Token cost is the small line item.**

Plus: **indemnity, compliance, harness** — often dispositive

NET

Enterprises don't pay the premium because they're bad at arithmetic — they pay it because the arithmetic they're doing has a different denominator, and because f enters additively while m divides. Where open weights genuinely win is a real and sizeable quadrant, but it's **bounded, high-volume work**, not general reasoning. The overpayment that actually exists is **failure to segment traffic**, and it shows up inside the proprietary cohort as often as across the closed/open line.

Notes & clarifications

Everything that is a choice rather than a measurement, stated plainly

Everything in this document that is a **choice rather than a measurement**, stated plainly, plus what changed in this edition and why.

Change log

HTML and PDF · October 2026 market refresh REV 2.2

Every market input was re-sourced on 20 October 2026; the derived arithmetic (drivers 1–3, the break-even identity, the 4.0% bar) is unchanged. (1) **Capability gap** — the Artificial Analysis Intelligence Index is now v4.3.2; the best open-weights model scores 46 against 58 for the best proprietary model, so the ratio quoted in the abstract and driver 4 falls from ~87% to ~79%. Driver 4 now marks both points: the wider gap puts open weights below the knee, strengthening rather than weakening the threshold argument. (2) **Reference prices** (driver 5) — list output prices have fallen: frontier flagship $75 → $50 per 1M tokens, mid-tier proprietary $15 → $10, open-weights API $6 → $2. (3) **Self-host cost** — 8×H200 rental raised from $26 to $36 per node-hour ($4.50/GPU-hour, low end of on-demand), taking fixed cost from $1.81M to $2.16M a year. Net effect on driver 5: self-hosting still beats a flagship price almost at once (3.8% → 6.9% utilisation), beats a mid-tier price only at the top of the realistic band (19.1% → 34.3%), and **no longer beats a cheap open-weights API at any utilisation** (was 47.9%). Lane 1 and the Net read are reworded to match. (4) **Lane 2** — "same index, a fifth the cost" is not supported by current data and is replaced with the measured comparison. (5) **Spend band** — unchanged, but Note 3 now cites reported averages and states that the band is the heavy-use tail.

HTML and PDF · revision in the document title REV 2.1.5

The revision number now appears in the document title — "Why Enterprises Pay the Frontier Premium · REV 2.1.5" — so it shows in the browser tab, in bookmarks, and in the PDF's document properties, not only in the tag beside the heading. The PDF was reprinted from this page and its footer carries the same revision. **No analytical content changed.**

HTML only · Download PDF links to the published PDF REV 2.1.4

The **Download PDF** button now downloads frontier-premium-economics.pdf, the PDF file published in the same folder as this page, instead of a copy stored inside the page. That copy was the older 9-page REV 2.1 edition, so the button now always gives whatever PDF is published alongside the page, and the page file shrinks from about 660 KB to about 370 KB. The PDF is printed from this page and its footer carries the same revision number. **No analytical content changed.**

HTML only · larger text · chart placement fix REV 2.1.3

All text enlarged about 15% for on-screen reading — body copy 12 px → 14 px, with headings, tables, chips, and chart labels scaled in proportion; chart areas made slightly taller to keep the larger labels clear. **Fixed** a display error in this page's break-even chart (driver 2): the 5% operating point and the shaded spend band were drawn at half their true x-position — the point at 2.5% and the band at 1.2–2.9% instead of 5% and 2.4–5.8% — because that chart's x-axis counts in 0.5-point steps. Both now sit correctly, matching the PDF, and the point's label has moved to open space with a leader line. The stated figures (4.0% break-even and the rest) were always correct; only the drawing was off. **No analytical content changed.** The downloadable PDF was never affected and remains REV 2.1.

HTML only · Download PDF button REV 2.1.2

Added a **Download PDF** button beside the theme switch. It saves the laid-out PDF edition (REV 2.1, 9 pages), which is embedded in this page so the button works offline and needs no second file. The browser's own Print → Save as PDF still works and always prints in the light theme, but its layout follows the screen; the laid-out edition is the one to share. **No analytical content changed.**

HTML only · light / dark theme REV 2.1.1

Added a Light / Dark / Auto theme control to the header. The page opens in Light by default; Dark switches every surface, chart, table, and chip to a dark palette; Auto follows the device setting. The choice is remembered in this browser, and printing always uses the light theme. **No analytical content changed** — every figure, claim, and note is identical to REV 2.1, which is why this is a patch-level revision. The PDF is unaffected and remains REV 2.1.

Post-review corrections · items 1–4 REV 2.1

Four corrections from an external arithmetic and consistency review, applied to both formats. (1) **Net read** — the withdrawn calibration-inversion driver had re-entered the summary table with a HIGH tag; row removed, driver count corrected to eight, and the hallucination-rate source line dropped since no figure in the body uses it. (2) **Driver 1** — the N=100 completion multiplier for 97%→99% is 7.7× (0.366 / 0.048), not 7.4×. (3) **Abstract, Method** — "reliability figures are third-party reported" contradicted Note 2; now states that per-step reliability values are illustrative parameters. (4) **Abstract, verdict, driver 1** — "clears the bar on any workload of length" replaced with the quantified claim: the two-point edge yields a completion-ratio gain of 2.1% at N=1, 4.2% at N=2, 22.6% at N=10, so it clears the 4.0% break-even from N=2 onward; the bridge from completion probability to m (accepted output, limit 4) is now stated explicitly.

Substitution modes · Mode A second-order claim CORRECTED

The previous edition stated that reducing headcount raises model spend as a percentage of loaded labour "mechanically." That is false. Mode A retains the same per-seat spend and the same per-seat throughput, so headcount N/m scales numerator and denominator together and f is unchanged at S/L. Replaced with the weaker and correct claim: the absolute AI budget shrinks with headcount, which can make a high-leverage line item look small and invite under-investment — a perception risk, not an arithmetic one.

Labour denominator · plain-language walkthrough NEW

Added a term-by-term reading of (1 + f) / m, a worked single-seat example in dollars and features, an elasticity table, the closed-form break-even E = (f_F - f_0) / (1 + f_0), and an explicit statement of the identity's limits.

Labour denominator · spend band REVISED

Model spend raised from 1–2% to 2.4–5.8% of loaded labour ($500–1,200/seat/month). The earlier band was anchored on light interactive use and understated saturated agentic sessions. The revision strengthens rather than weakens the conclusion — break-even rises only to 4.0%.

Labour denominator · presentation REFRAMED

Replaced the break-even bar chart with the (1 + f) / m unit-cost identity and its isoclines. The bar chart answered "what lift repays the spend" against an absolute baseline; the isoclines answer the decision-relevant question, which is the lift required relative to the open-weights alternative.

Substitution modes NEW

Added as a distinct driver. Headcount reduction and output expansion are algebraically identical on cost per unit of output and differ only in where value is booked and whether it is bounded.

Jevons drift NEW

Noted that total spend rises as unit cost falls, and that percentage-of-labour eventually stops being the correct accounting frame.

Prior-revision corrections CARRIED

Corrections from the previous edition remain in force: the provenance-laundered calibration attribution and the invalid hallucination-rate denominator. The calibration-inversion driver remains withdrawn rather than reversed.

Notes & clarifications

1 **Throughput multipliers are parameters, not measurements** LOW

m = 1.20 / 1.45 / 1.55 are chosen to illustrate the shape of the unit-cost relation. They set the size of the gain, not its direction. Break-even is independent of m entirely — it is (1 + f_frontier) / (1 + f_open) — so the 4.0% figure survives any revision to them. First-party measured throughput would rank above these and should replace them.

2 **p^N is an upper bound on decay, not a forecast** HIGH

The independence assumption overstates compounding: real agentic failures correlate, and recovery, retry, and human checkpoints truncate the chain. The curves describe the shape of the effect and the direction of the reliability premium. Specific completion rates are illustrative.

3 **Spend band is inferred, not surveyed** MED

$500–1,200/seat/month is reasoned from agentic usage patterns rather than measured across a population, and it describes the heavy-use tail, not the typical seat. For calibration: Anthropic reports average Claude Code spend of about $150–250 per developer per month, with 90% of users under $30 per active day; a Gartner survey reported in June 2026 found about a quarter of technology leaders spending $200–500 per developer per month and about 6% spending over $2,000. A lower typical spend only lowers the break-even bar (E ≈ Δf), so the band is the conservative choice. The right tail is fat: parallel-session users can exceed it materially.

4 **Self-host TCO is dominated by one assumption** LOW

5,000 sustained output tok/s per node for a ~100B-class MoE governs every crossover in driver 5. Halve it and each crossover doubles. Node rental ($4.50/GPU-hour, the low end of October 2026 on-demand H200 pricing; committed contracts run lower) and loaded FTE cost are market estimates. The qualitative ordering survives; the specific percentages do not.

5 **Threshold curve is illustrative on curvature** MED

The sigmoid's steepness and midpoint are chosen, not fitted. The direction — that value transfer is convex rather than linear on gated work — is the load-bearing claim.

6 **No composite averaging across heterogeneous rows** HIGH

The 79% figure (and June's 87%) is quoted only as the thing being critiqued. It is a ratio of means over ten evals and is not used here as an input to any calculation.

7 **Reference prices are output-only** MED

API prices are list output prices on 20 October 2026 and exclude input tokens, cache reads, and the verbosity tax on output-heavy models. Folding verbosity in shifts effective frontier cost upward and moves the self-hosting crossovers left.

8 **Commercial terms are excluded** HIGH

Project-level discounts, committed-use pricing, and enterprise volume agreements materially compress the frontier premium and appear in no figure here. Every stated premium is therefore an upper bound on what a negotiating enterprise actually pays.

Sources. Reliability compounding and the unit-cost identity are derived arithmetic. Index and cost-per-task figures from the Artificial Analysis Intelligence Index v4.3.2, live 20 October 2026 (June comparisons: v4.0, 9 June 2026); list prices and H200 rental from provider price pages, same date. Every modelled parameter, limitation, and exclusion in this document — including throughput multipliers, the spend band, self-hosting assumptions, and omitted commercial terms — is itemised with confidence tags on the **Notes & clarifications** page.